

Putting outlet, platform and feed negativity on one scale: a measurement design for SP1

Governing question: how much negativity, and what kind, do Danish news outlets publish, do platforms carry, and do individual feeds contain, and what has to be true of the instrument before those three quantities can be compared?

Research proposal for the PhD position in SP1, Navigating Negativity: What is a Healthy News Diet? · Centre for Journalism, University of Southern Denmark · Asad Ikram · September 2026

STARTING POINT

SP1 asks how the amount and character of negativity differ across news outlets, social media platforms and individual news diets. Comparing one measure across three text types that differ in length, register and composition is the hard part of that question, and this field knows it well: automated tone and sentiment measures lose accuracy when moved between domains and genres (van Atteveldt, van der Velden and Boukes 2021; Boukes et al. 2020), and Chan et al. (2021) found the dominant latent factor across dictionary scores on news tracks article length almost perfectly, which is exactly what separates an article from a post from a caption. Any validation SP1 undertakes will meet this. My proposal is to make that validation systematic and to carry its results forward, so a reported difference between an outlet and a feed can be attributed to the sources rather than to the ruler. This is not only SP1's concern. SP3's exposure variable is built from SP1's numbers, and Scharnow and Bachl (2016) showed that content-analysis error of this kind attenuates linkage estimates and can manufacture minimal effects. Getting the measure right is what makes the downstream findings defensible.

CONSTRUCT AND UNIT

Negativity is not one quantity. Lengauer, Esser and Berganza (2012), whose phrase "amount and character" the project uses, separate negativity from confrontation, frame-related from actor-related negativity, and non-directional from directional tone; the DOCA conventions distinguish negative tone toward actors, overall tonality, pessimistic outlook, conflict-centredness and attributions of incapability. Soroka, Young and Balmas (2015) show valence and the discrete emotions of fear and anger are empirically distinct, so amount and character are separate estimands and one scalar cannot carry both. The primary coding unit is the document, because a feed contains many items with no identifiable actor, and actor-level coding in the tradition of Hopmann, de Vreese and Albæk (2011) is nested inside it where the text supports it. The share of items eligible for actor-level coding is reported per source, since it will differ systematically between an article and a caption, and that difference is a finding rather than a nuisance. The aggregation rule from document to the person-day quantity the project needs is fixed in advance with its denominator stated. I treat the negativity bias as an open question rather than a premise, following Andersen and Shehata's (2025) preregistered finding of a preference for positive news.

WHAT THE DONATED DATA CAN AND CANNOT SAY

A download package is not a record of what a platform showed someone. Some packages carry viewed-content logs, Instagram's viewed posts and videos among them, while others record only what the account posted, liked, shared or followed. The distinction decides what the feed layer can measure, because engaged negativity and encountered negativity differ by exactly the selective attention this programme studies, and an exposure variable built from engagement would carry the phenomenon inside the predictor. WP2 therefore establishes, platform by platform, which fields correspond to encounter, which to engagement and which to neither, what retention window each covers, and what share resolves to analysable text. What the feed layer supports is declared before it is used, and the person-day measure handed to SP3 is labelled for what it actually is.

WORK PACKAGES

WP1, PAPER 1. A DANISH NEGATIVITY INSTRUMENT, VALIDATED PER SOURCE. A codebook from Lengauer's axes and the DOCA variables; a gold standard double-coded by Danish-native coders with an iterated coding list; Krippendorff's alpha reported per construct and per source. A multilingual encoder is fine-tuned and benchmarked against AFINN, SENTIDA, the DaNLP tone model and an LLM pass, since on Danish data the lexical tools fall away across the formal-to-informal transition this project depends on (AFINN 0.68 on Europarl to 0.48 on Twitter) while transformer tone models hold better. Automated tone measurement on Danish news is established here already, in van Dalen, de Vreese and Albæk's (2017) time series on economic coverage. What I have not found is a Danish tone measure validated on news and social text within one study, which is what a cross-source comparison requires. *Limitation:* the gold standard three years permits will bound the ceiling on rare constructs such as attributions of misconduct.

WP2, PAPER 2. THE DONATION HUB AS A MEASURING INSTRUMENT. A schema-drift detection and repair layer over download packages, with golden fixtures per platform and per locale, versioned parsers stamped onto every record, and canary donations from project-owned accounts, because package structure changes with the account's language setting and Danish donors' phones will split between Danish and English. A content-resolution layer, because donated entries are identifiers and links rather than text and no classifier can read them until they are resolved at scale, which Hase et al. (2024) name as the blocker for content analysis of donated feeds. Selective redaction with a public-entity allowlist, since blanket handle-redaction would erase the outlet identity the comparison depends on. On-device extraction with two-stage consent. Every funnel step is logged, so the hub yields an evaluation of assisted physical donation as a method. *Limitation:* resolution fails for deleted and restricted items, biasing the recovered feed toward what is still live; resolution rate is reported by platform, item age and outlet rather than assumed away.

WP3, PAPER 3. EQUIVALENCE AND CORRECTION. The design is multitrait-multimethod: every sampled unit carries scores from mechanically different methods, the fine-tuned encoder, two dictionaries, an LLM pass and a double-coded human label, which is the indicator set an equivalence analysis needs. A generalizability study decomposes score variance over source, instrument, coder and format as crossed facets. Differential functioning is tested by conditioning on the human codes as an external criterion rather than on a latent estimate. Where sources are not equivalent, a per-source misclassification model is estimated against that criterion and propagated into every aggregate SP2 and SP3 consume, following Egami et al. (2023). Length enters as a modelled facet. *Limitation:* genuine content differences between sources and instrument non-transfer are confounded on naturally occurring text, so identification rests on

the human criterion holding constant across sources; a transplantation check presenting matched content in each format tests exactly that.

WP4, PAPER 4. THE COMPARISON, CORRECTED. The descriptive SP1 comparison is delivered twice. First in year one on outlet and platform text with the instrument as built, which gives the project an early answer rather than a promise. Then again once WP3's correction exists, which shows what the correction changes and by how much. The second pass adds the feed layer and hands SP3 a person-day measure with a stated error model, shaped for joining as Hopmann, Vliegenthart, de Vreese and Albæk (2010) shaped daily content data for rolling cross-section survey data. *Limitation:* donated feeds are a self-selected sample of a self-selected platform population, benchmarked against SP3's panel on observable characteristics.

SAMPLING, ETHICS AND LANGUAGE

The outlet frame is the Danish national and regional press through a media archive with random-day sampling, extended to alternative outlets. Donated feeds are a non-probability sample; expected yield is planned from published donation studies rather than assumed, and the donor journey is costed in wall-clock time with package delivery latency inside it, because absorbing that latency is what a staffed hub is for. Reliability is reported before performance, alpha per construct per source alongside macro-F1, and an LLM pass is validated against the Danish gold standard rather than trusted. Ethics is designed in: a data protection impact assessment before any donation, two-stage consent with the donor inspecting the extracted preview before releasing it, on-device extraction so the raw package never leaves the phone, selective redaction validated against a re-identification assessment, explicit handling of third parties and of donors who were minors when the account was used, and storage under SDU governance. Language is a third axis of non-transfer alongside domain and genre and enters the model as a facet. I do not speak Danish: the codebook and the reliability targets are mine to develop with the team, native-speaker coders produce the gold standard and I work inside their disagreements, which is where a codebook is actually written, and I would begin Danish tuition on arrival.

THREE YEARS, AND WHAT IT PRODUCES

Year 1: codebook, gold standard, WP1 instrument, parser and resolution layer against public fixtures, and the first descriptive comparison on outlet and platform text. Year 2: hub fielded; WP3. Year 3: the corrected comparison, the SP3 handover, thesis. The hub is scheduled from year one and not made contingent, because it is the part of SP1 that cannot be bought in; if donor yield falls short, the feed layer narrows to a smaller validated sample and the hub evaluation stands on the funnel data regardless. *To research:* a Danish negativity instrument validated separately per source; an equivalence and correction treatment of an automated content-analysis measure; and an evaluation of assisted physical data donation, which I have not found published. *To practice:* an open codebook, parsing and resolution tooling and a reporting standard another centre can run; outlet-level findings Danish editors can act on; and a negativity estimate whose error is stated, which is the difference between advice on news diets that can be defended and advice that merely sounds careful.

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